WENZHAN LI

Embodied AI ▪ World-Action Models ▪ Robot Learning

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Research Profile

Research in embodied AI, focused on world-action models (WAMs), robot learning, and adaptation. Experience leading research projects and developing methods, research codebases, and experiments for vision-language-action (VLA) policy refinement and cross-domain prediction. Additional work spans sim-to-real data generation and robotics algorithm engineering at RoboSense.

Education

Sun Yat-sen University, Shenzhen Campus Sep. 2026 – Present
Ph.D. student in Artificial Intelligence
School of Electronics and Communication Engineering
Advisor: Prof. Yulan Guo

Joint doctoral training at **Shenzhen Loop Area Institute (SLAI)**
Center for Embodied Artificial Intelligence and Computer Vision (EACV)

Tsinghua University Sep. 2020 – Jun. 2023
M.Eng. in Mechanical Engineering, Intelligent Manufacturing
Shenzhen International Graduate School GPA: 3.71/4.0

Xi'an Jiaotong University Sep. 2016 – Jun. 2020
B.Eng. in Mechanical Engineering GPA: 3.35/4.0

Selected Publications

*Co-first authorship of Wenzhan Li.

Trust-Calibrated VLA Policy Post-Training with Evolving Imagination
Wenzhan Li, Yiran Qin, Heng Zhou, Huirui Wang, Yulan Guo, Ruimao Zhang.
CoRL 2026 | Accepted MERL • Sole first author

PAWN: Preference-Aware Semantic NFT for Online Reinforcement Learning of Flow-based VLAs
Haomin Zuo, **Wenzhan Li***, Ruimao Zhang, Yulan Guo.
CoRL 2026 | Accepted Co-first author, listed second

Building Scalable Real-World Robot Data Generation via Compositional Simulation
Yiran Qin, Jiahua Ma, Li Kang, **Wenzhan Li***, Yihang Jiao, Xin Wen, Xiufeng Song, Heng Zhou, Zhemeng Zhang, Zhenfei Yin, Xihui Liu, Philip Torr, Ruimao Zhang.
CoRL 2026 | Accepted ComSim • Co-first author, listed fourth

Dual-Alignment Domain Adaptation for Pedestrian Trajectory Prediction
Wenzhan Li*, Fuhao Li, Xinghui Jing, Pingfa Feng, Long Zeng.
IEEE Robotics and Automation Letters, 2024 DADA • Co-first author, listed first
Vol. 9, no. 12, pp. 10962–10969. DOI: 10.1109/LRA.2024.3481831

Selected Research: Accepted and Published

MERL: Reliable World-Model-Based VLA Refinement
CoRL 2026

Project lead and primary contributor

- Led method design, research codebase development, experimental design and execution, and manuscript writing; coordinated project milestones, resources, and technical quality review.
- Designed policy refinement with an evolving world model and reward proxy. Two-level trust calibration controls imagined-data ratios and rollout horizons, then filters and weights imagined action chunks to limit unreliable model-generated experience.
- On LIBERO-PRO simulation, achieves **79.7% average success vs. 77.2%** for model-free RL with the same policy initialization and online training budget. Real-robot validation uses offline refinement followed by deployment for inference.

PAWN: Preference-Guided RL for Flow-Based VLAs
CoRL 2026

Project lead

- Co-designed chunk-level scoring and same-prefix continuation ranking; developed the theoretical formulation and proofs for the chunk-level scorer used in preference-based policy refinement.
- Coordinated project execution, experimental infrastructure, and resources; contributed selected experiments and manuscript sections.
- With $\pi_{0.5}$ on LIBERO simulation, reaches **97.9% average success vs. 94.0%** for π-StepNFT under the same rollout budget. The approach avoids explicit action-likelihood estimation; real-robot adaptation is performed offline.

ComSim: Action-Aligned Sim-to-Real Data Generation
CoRL 2026

Team member • Neural simulator development

- Implemented the neural simulator's core action-conditioning and visual-conditioning components within the team's real-sim-real data-generation framework.
- Contributed selected experiments and manuscript sections. The paper evaluates policy success and generalization on real-robot tabletop tasks with limited real demonstrations, targeting the visual appearance gap while retaining simulated action trajectories.

DADA: Cross-Domain Pedestrian Trajectory Prediction
RA-L 2024

Primary contributor

- Designed the overall dual-alignment method and developed the core algorithms and pipeline: data-level adversarial trajectory transfer with temporal convolution and graph attention, coupled with feature-level adversarial learning of domain-invariant representations.
- Contributed experimental design, execution, and manuscript sections. Evaluation covers **20 source-to-target tasks on ETH/UCY** and six prediction backbones, with component ablations and lower average final displacement error than the unadapted baselines.

Research Appointment

Sun Yat-sen University | Research Assistant

Contracted through The Chinese University of Hong Kong, Shenzhen.

Research on world-model-based manipulation refinement and sim-to-real data generation.

Current Research: World-Action Models

Three concurrent research projects initiated in July 2026, focused on learning and adaptation in robotic manipulation.

Experience Reuse for Robot Learning Jul. 2026 – Present

- **Project lead.** Directing research on experience reuse for reliable robotic manipulation; responsible for concept and study design, resource coordination, and contributions to experimental execution and manuscript preparation.

World-Action Model Adaptation Jul. 2026 – Present

- **Project lead and primary contributor.** Leading research on WAM post-training and adaptation through interaction; responsible for concept development, experimental design and execution, resource coordination, and contributions to manuscript preparation.

World and Action Representation Learning Jul. 2026 – Present

- **Project lead and primary contributor.** Investigating world representations for useful and efficient action learning; responsible for concept development, experimental design and execution, resource coordination, and contributions to manuscript preparation.

Industry Experience

RoboSense | Robotics Algorithm Engineer Sep. 2023 – Jul. 2025

AI Group, Decision-making Team • Shenzhen, China

- Built automated annotation tools and coordinated manual annotation for model training; fine-tuned pretrained models and implemented output post-processing.
- Verified online/offline model consistency, implemented algorithm modules, and resolved field-test cases through iterative improvements to runtime processing logic.

Pudu Robotics | Intern Sep. 2020 – Jun. 2021

- Developed and evaluated a YOLOv3-based detector with BiFPN for restaurant scenes; tuned multi-scale feature fusion and network parameters for detection accuracy and inference speed.

Programming: Python, C++.
Tools & platforms: PyTorch, MySQL, ROS 2, Baidu Apollo.

Collaborative Research

Agentic 3D Indoor Scene Generation Ongoing

SLAI EACV project • Team member

Project faculty: Profs. Ping Tan, Yizhou Yu, Yulan Guo, and Guanbin Li.

- Developing the agent orchestration (harness) framework and researching and implementing memory mechanisms within a team project on indoor 3D scene generation for embodied AI.

Future Research Interests

Exploratory interest in **collective self-improvement for heterogeneous robots** across embodiments, environments, and tasks. Interested in how a shared knowledge backbone and robot-specific knowledge bases could support continual adaptation and collaborative improvement.